Data Science

CSE558

Recap

- Likelihood Estimation

Overview

- Aposteriori Estimation

Conditional Probability

Posterior

- $P(H \mid E)$: Probability that your friend have the disease given he tested positive
- $P(H \mid E) = P(E \cap H)/P(E)$

Prior

- $P(H)$: Probability of having the disease

Likelihood

- $P(E \mid H)$: Probability that he would test positive given he had the disease

Marginal

- $P(E)$: Probability of testing positive
- $P(E) = P(H)P(E \mid H) + P(\sim H)P(E \mid \sim H)$
- Probability of having the disease and correctly tested positive +
- Probability of not having the disease and falsely tested positive

Conditional Probability

Bayes' Rule

$$\mathbf{P}(H|E) = \frac{\mathbf{P}(E|H) \cdot \mathbf{P}(H)}{\mathbf{P}(E)}$$

$$= \frac{\mathbf{P}(E|H) \cdot \mathbf{P}(H)}{\mathbf{P}(H) \cdot \mathbf{P}(E|H) + \mathbf{P}(\sim H) \cdot \mathbf{P}(E|\sim H)}$$

Bayes' Rule

Prior

P(H): Probability of having the disease = 0.001

Likelihood

P(E | H): Probability your friend would test positive given he had disease = 0.99

Marginal

P(E): Probability of testing positive

$P(E) = P(H)P(E \mid H) + P(\sim H)P(E \mid \sim H) = 0.001 \cdot 0.99 + 0.999 \cdot 0.01$

P(H | E) = 0.09 = 9%

True potential is yet to be determined!

Bayes' Rule

Update your understanding!

<u>Prior</u>

P(H): Probability of having the disease = 0.09

<u>Likelihood</u>

P(E | H): Probability your friend would test positive given he had disease = 0.99

<u>Marginal</u>

P(E):Probability of testing positive

$P(E) = P(H)P(E \mid H) + P(\sim H)P(E \mid \sim H) = 0.001 \cdot 0.99 + 0.999 \cdot 0.01$

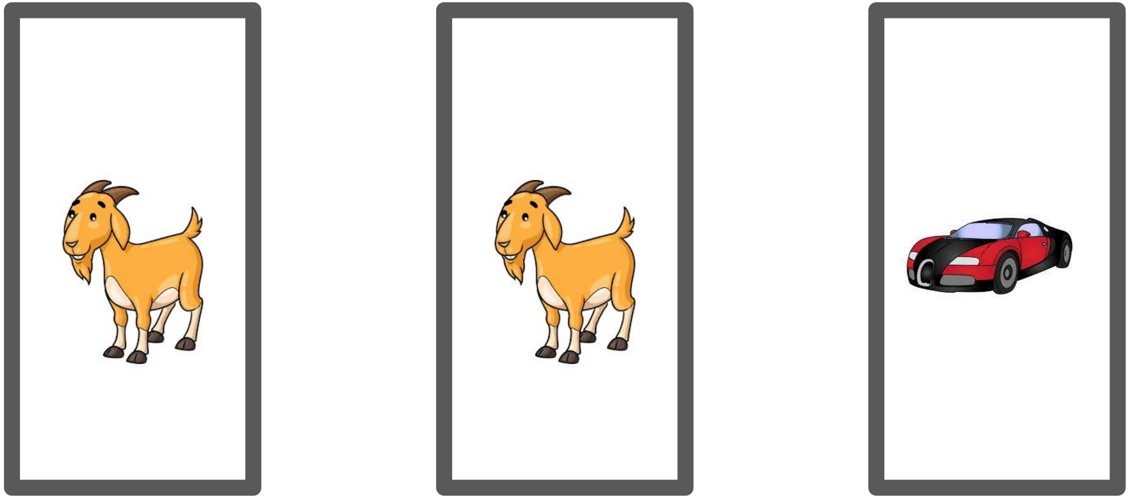
P(H | E) = 0.90 = 90%

Game

Go to: bit.ly/ds-game

Game

Go to: bit.ly/ds-game



Rule/About Game

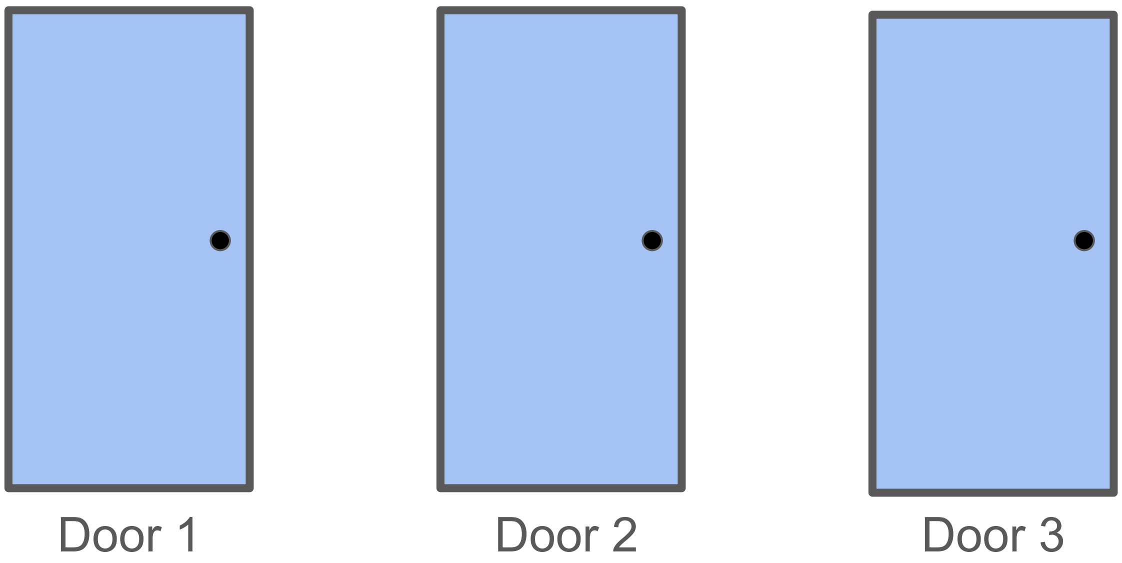
1. You select a door.
2. Monty opens a door with a goat, from the remaining two non-selected doors.
3. You get a chance to change your selection, before the next door is revealed.

Note: Rewards behind doors are fixed (even before starting the game!)

Game

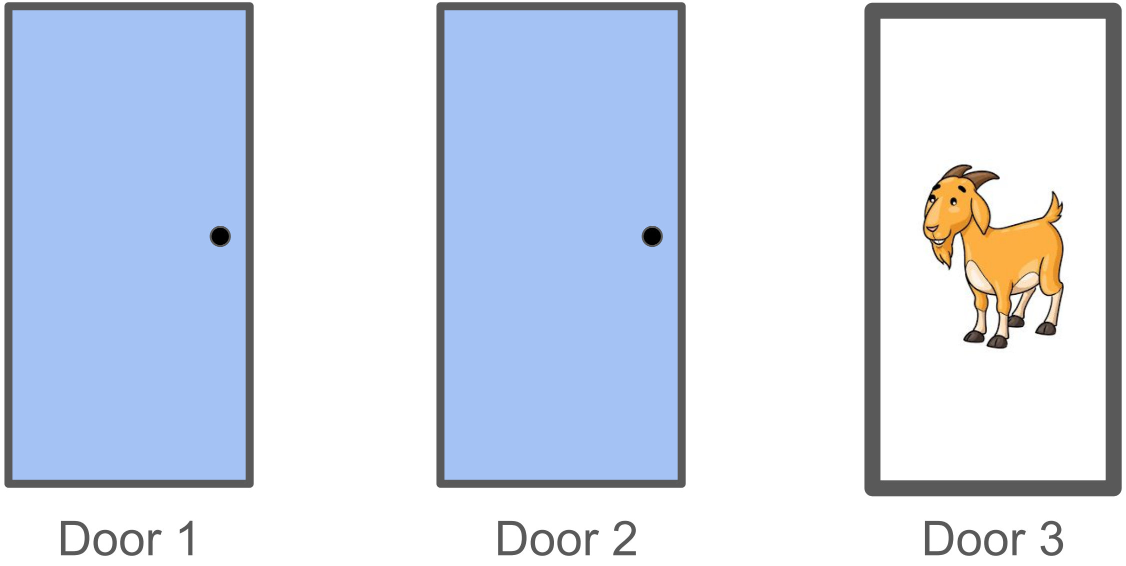
You choose a door

Go to: bit.ly/ds-game



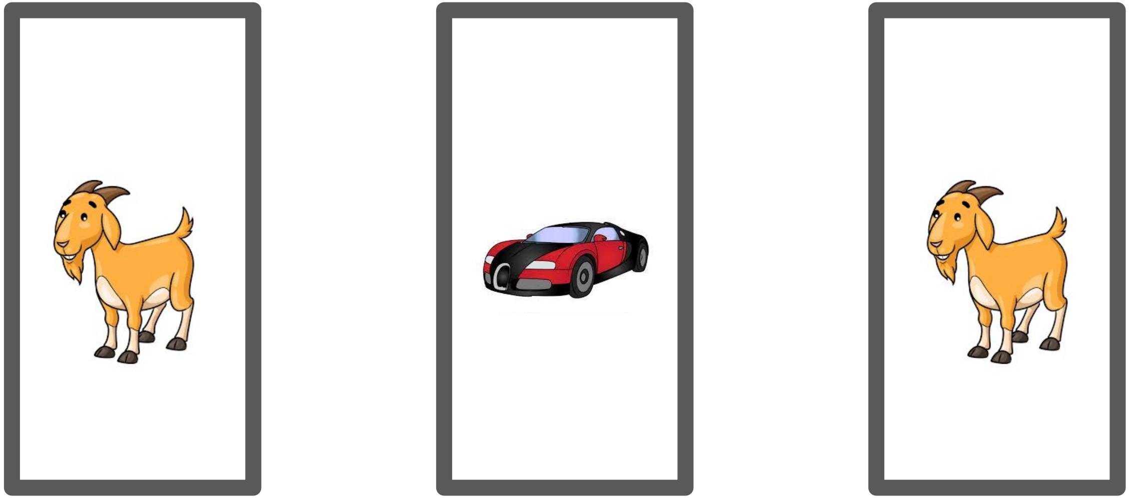
Game

Monty opens a door that you have not chosen and has goat behind it.
Now you have the option of changing your choice.



Game (Monty Hall Problem)

Monty opens the remaining doors.



Door 1 Door 2 Door 3

Understand using Bayes' Rule

H: Door that has the car (H = 1, 2, 3) & let, you select door1.

E: Door that Monty reveals (E = 1, 2, 3)

Here, E = 3

Not changing choice.

Your hope is that the following conditional probability would be high!

$$\mathbf{P}(H = 1 | E = 3) = \frac{\mathbf{P}(E = 3 | H = 1) \cdot \mathbf{P}(H = 1)}{\mathbf{P}(E = 3)}$$

Not Changing Choice

$$\mathbf{P}(H=1|E=3) = \frac{\mathbf{P}(E=3|H=1)\cdot\mathbf{P}(H=1)}{\mathbf{P}(E=3)}$$

H: Door that has the car (H = 1, 2, 3) & let, you select door1.

E: Door that Monty reveals (E = 1, 2, 3)

Say, E = 3

P(E=3 | H=1) is the probability of opening door3, if the car is behind door1 = ½

P(E=3 | H=2) is the probability of opening door3, if the car is behind door2 = 1

P(E=3 | H=3) is the probability of opening door3, if the car is behind door3 = 0

Probability of the car behind one of the door is P(H=1) = P(H=2) = P(H=3) = ⅓

P(E=3) = P(H=1)·P(E=3 | H=1) + P(H=2)·P(E=3 | H=2) + P(H=3)·P(E=3 | H=3)

= ⅓·(½ + 1 + 0) = ½.

P(H=1 | E=3) is the probability that the car is behind door1 given Monty opened door3 = (½·⅓)/½ = ⅓.

Changed Choice

H: Door that has the car (H = 1, 2, 3) & let, you select door1.

E: Door that Monty reveals (E = 1, 2, 3)

Say, E = 3

Changed, selection to door2;

Now you hope that the following conditional probability would be high!

$$\mathbf{P}(H = 2|E = 3) = \frac{\mathbf{P}(E = 3|H = 2) \cdot \mathbf{P}(H = 2)}{\mathbf{P}(E = 3)}$$

Changed Choice

$$\mathbf{P}(H = 2|E = 3) = \frac{\mathbf{P}(E = 3|H = 2) \cdot \mathbf{P}(H = 2)}{\mathbf{P}(E = 3)}$$

H: Door that has the car (H = 1, 2, 3) & let, you select door1.

E: Door that Monty reveals (E = 1, 2, 3)

Say, E = 3

Changed, selection to door2;

$P(E=3 \mid H=1) = \frac{1}{2}$, $P(E=3 \mid H=2) = 1$ & $P(E=3 \mid H=3) = 0$

$P(H=2) = \frac{1}{3}$

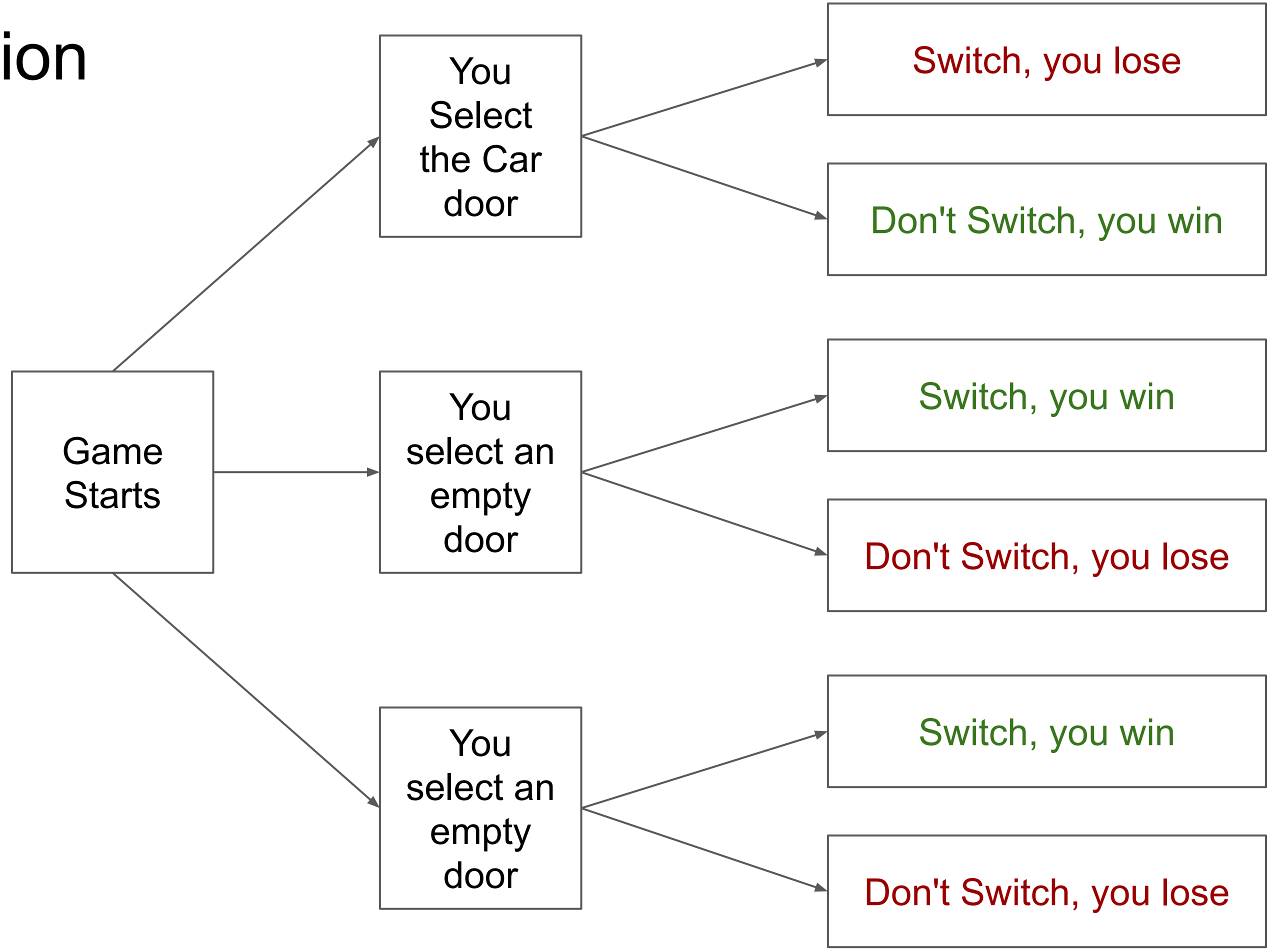
$P(E=3) = P(H=1) \cdot P(E=3 \mid H=1) + P(H=2) \cdot P(E=3 \mid H=2) + P(H=3) \cdot P(E=3 \mid H=3)$

$= \frac{1}{3} \cdot (\frac{1}{2} + 1 + 0) = \frac{1}{2}$.

$P(H=2 \mid E=3) = (\frac{1}{3})/\frac{1}{2} = \frac{2}{3}$.

Changing the choice will increase your chance of winning!

Intuition


Game Starts
You Select the Car door
You select an empty door
You select an empty door
Switch, you lose
Don't Switch, you win
Switch, you win
Don't Switch, you lose
Switch, you win
Don't Switch, you lose

Maximum a Posteriori

Sample

Distribution of Age

Start with Assumption **& Prior**

Find parameter that fits sample, **given a Prior**

Maximum a Posteriori

Let there are random samples $\mathbf{X}_1$, $\mathbf{X}_2$, …, $\mathbf{X}_n$, whose distribution depends on some parameter ϴ. Let P(ϴ) be the prior on ϴ.

- Bernoulli (*p*)
- Binomial (*p*)
- Geometric (*p*)
- Normal (μ, σ)

The problem is to find the parameter ϴ that maximizes the likelihood of of the samples and the prior on ϴ.

Maximum a Posteriori (MAP)

Let there are random samples $\mathbf{D} = \{\mathbf{X}_1, \mathbf{X}_2, \ldots, \mathbf{X}_n\}$, whose distribution depends on some parameter ϴ. Let P(ϴ) be the prior on ϴ.

$$\mathbf{P}(\theta|\mathbf{D}) = \frac{\mathbf{P}(\mathbf{D}|\theta)\mathbf{P}(\theta)}{\mathbf{P}(\mathbf{D})}$$

$$\text{argmax}_\theta \mathbf{P}(\theta|\mathbf{D}) = \text{argmax}_\theta \frac{\mathbf{P}(\mathbf{D}|\theta)\mathbf{P}(\theta)}{\mathbf{P}(\mathbf{D})}$$

$$\text{argmax}_\theta \mathbf{P}(\theta|\mathbf{D}) \approx \text{argmax}_\theta \mathbf{P}(\mathbf{D}|\theta)\mathbf{P}(\theta)$$

MAP of Normal Distribution

Let $\mathbf{D}$ = $\{\mathbf{X}_1, \mathbf{X}_2, \ldots, \mathbf{X}_n\}$ are independent random samples from a normal distribution. Let the prior be ϴ ~ N(μ, 1). What is maximum a posteriori?

$$\theta_{MAP} \in \operatorname{argmax}_\theta \mathbf{P}(\theta|\mathbf{D})$$

$$\theta_{MAP} \in \operatorname{argmax}_\theta \log\left(\mathbf{P}(\mathbf{D}|\theta)\right) + \log\left(\mathbf{P}(\theta)\right)$$

$$\frac{\partial \log\left(\mathbf{P}(\mathbf{D}|\theta)\right)}{\partial\theta} + \frac{\partial \log\left(\mathbf{P}(\theta)\right)}{\partial\theta}$$

$$\frac{1}{\sigma^2}\sum_{i=1}^{n}(\mathbf{X}_i - \theta) - (\theta - \mu)$$

$$\theta = \frac{\frac{1}{\sigma^2}\sum_i \mathbf{X}_i + \mu}{\frac{n}{\sigma^2} + 1} = \frac{\sum_i \mathbf{X}_i + \sigma^2\mu}{n + \sigma^2}$$

Classwork

1. Let you get a sample population with their ages.

Age	20	30	40	50	60
Frequency	2	4	6	5	3

What are the parameters of a normal distribution that fits the above sample given $\Theta \sim N(38, 400)$.

Properties

1. Avoids overfitting.

2. When $n \rightarrow \infty$ then $\Theta_{MAP} \rightarrow \Theta_{MLE}$